effective team and can organize the project and make firm, wise decisions at every major juncture.

Out came the shotgun again. If you want your project to continue, NSF's Isaacson told us, you must find that director and learn to work with him like a football team works with a great coach or an orchestra with a great conductor.

We were lucky. In the midst of our search, Robbie Vogt got fired.

Vogt, a brilliant, strong-willed experimental physicist, had directed projects to construct and operate scientific instruments on spacecraft, had directed the construction of a huge millimeter-wavelength astronomical interferometer, and had reorganized the scientific research environment of NASA's Jet Propulsion Laboratory (which carries out most of the American planetary exploration program)—and he then had become Caltech's provost. As provost, though remarkably effective, Vogt battled vigorously with Caltech's president, Marvin Goldberger, over how to run Caltech—and after several years of battle, Goldberger fired him. Vogt was not temperamentally suited to working *under* others when he disagreed profoundly with their judgments; but on top, he was superb. He was just the director, the conductor, the coach that we needed. If anybody could mold us into a tightly knit team, he could.